JAISUBASRI KARTHIKEYAN
21PD14

Gender	Female
Date of Birth	17th March 2004
Languages known	English, Tamil, French
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66-N6, Ayyan Thottam, Maharishi Nagar, Rajsri Bhavan,
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ACADEMIC QUALIFICATION

Currently pursuing 5th year of Integrated M.Sc. Data Science at the Department of Applied Mathematics and Computational Sciences in PSG College of Technology, Coimbatore.

SKILL SET

Languages	Python, C++, R, SQL
Library	Sklearn, Keras, Scipy, Seaborn, Plotly, OpenCV ,Pytorch
Tools and Technology	SQL developer, Tableau, Power BI, Streamlit, Flask, Dialogflow, Azure (Azure Databricks, Azure Data Factory, Azure Data Lake Storage), PowerApp, Power Automate, TigerGraph

AREAS OF INTEREST

- Data structures and Algorithm
- Data Visualization and Analytics
- Natural Language Processing
- Machine and Deep learning

ACADEMIC RECORD

- **M.Sc. Data Science** — *2021-2026*
  PSG College of Technology, Coimbatore — **9.03 CGPA**
- **XII (Higher secondary, State Board)** — *2021*
  S.S.M Lakshmi Ammal Matric. Hr. Sec School, Komarapalayam — **97.51%**
- **X (SSLC, State board)** — *2019*
  S.S.M Lakshmi Ammal Matric. Hr. Sec School, Komarapalayam — **96.67%**

INDUSTRY BASED PROJECT EXPERIENCE

- **CLEO SOFTWARE SOLUTIONS |** Engineering Intern
  (May 2023 – June 2023)

The Chatbot project employs **Diaglowflow and GPT models** (ada, babbage, curie, davinci) for an intelligent conversational agent. Unique models are built for business, personal, private use. The bot is deployed as a mobile app with a website interface.

- **THOROGOOD ASSOCIATES INDIA PRIVATE LIMITED |** Data & AI Consultant
  (June 2024 – November 2024)

Cutlass System – Case Study & Business Assurance Project

The Cutlass System case study, utilizing various **Azure technologies** for data preprocessing and structuring data to handle exceptions and ensure alignment with business goals. Additionally, contributed to the development of the Project Health App, a web-based application integrated with a **Microsoft PowerApp, Microsoft PowerAutomate**, **SQL database**, designed to monitor and assess client project health, ensuring seamless data management and reporting.

- **INSTITUTE OF MATHEMATICAL SCIENCES |** Junior Researcher
  (December 2025 – May 2026)

Benchmarking Artificial Intelligence Models for Predicting Nuclear Receptor Activity

Conducted benchmarking of **ML, DL, and Transformer models** on Tox21 datasets (43 datasets, 18 receptors) to predict chemical bioactivity. Found **tree-based models** perform best on balanced data, while deep learning handles moderate imbalance better. Identified **class imbalance** and chemical diversity as key factors, with ~40% misclassified chemicals being structurally isolated. External validation showed strong generalization for receptors like AR and ER, supporting reliable AI models for toxicity screening.

NON - ACADEMIC PROJECTS

- **FORMULA 1 DRIVER PERFORMANCE** is a **time series model** using **XGBoost and Markov chains** to forecast Formula 1 driver finishing positions. Leveraged **TensorFlow, SkLearn, and Keras** for model development with **Python and SQL**. Built an interactive **Streamlit web app** and provided insights via **Power BI**. Enhanced accuracy through **feature engineering and hyperparameter tuning**, aiding sponsors in strategic decision-making.
- **FLOORPLANNING** is a *B-Tree-based floorplanning algorithm** inspired by *"Modern Floorplanning Based on B*-Tree and Fast Simulated Annealing" (IEEE)*. Implemented **simulated annealing** to optimize block placement, minimizing **wirelength and dead space** while managing **interconnect constraints**. Utilized **adaptive weight tuning, Python, and heuristic search** for efficient **VLSI physical design automation**.

ACADEMIC MINI PROJECTS

- **TUMOR TRACE PROGNOSIS** combines classical machine learning **(Logistic Regression, SVM, KNN)** with deep learning **(DenseNet, Inception)** to analyze mammograms for improved breast cancer detection. Using **Sklearn, Keras**, it aims to identify subtle patterns, enhance early diagnosis, and reduce mortality rates. The approach involves comparing diverse algorithms on platforms like Kaggle, developed with tools like VSCode.

- **ELECTRICAL SALES FORECASTING** is a web application developed using **Python**, utilizing the **streamlit libraries**. The project involves the analysis of real-time electrical sales data using Python prediction models, accompanied by the utilization of **Power BI** for interactive data visualization. The aim is to provide accurate sales forecasts to enhance decision-making in the electrical sector.

EXTRA-CURRICULAR ACTIVITIES AND ACHIEVEMENTS

- Secured 1st prize in DATA PREMIER LEAGUE hackathon conducted by PSG Institute
- Participated and won prizes in many events in and outside of my college and so recognized with Achiever award of batch 2021-2026 in Data Science
- Ranked 4th place in Science Olympiad conducted at national level by Indian Talent.
- Ranked 4th place in Olympiad conducted at national level by conquest IQ.
- Participated in the Guinness World Record and won Asia Book of Record in Bharatanatyam.

DECLARATION

I, JAISUBASRI K, do hereby confirm that the information given above is true to the best of my knowledge.

Place: Coimbatore
Date: 13/04/2026 (Jaisubasri K)